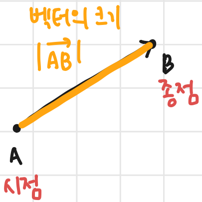
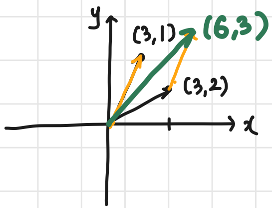
• 벡터 : 화살표, 점이다. : $\overrightarrow{AB}$



• 벡터의 덧셈

◦

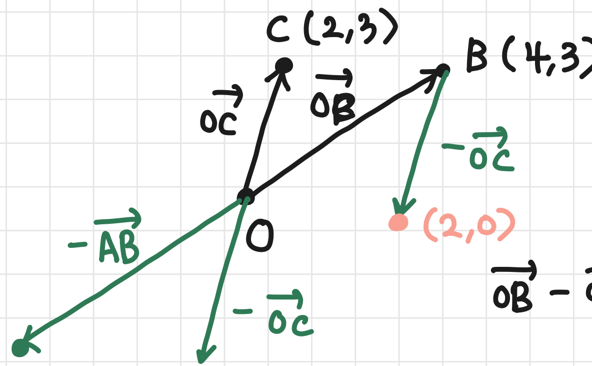


◦ 덧셈의 성질

(1) 교환법칙 : $\vec{a} + \vec{b} = \vec{b} + \vec{a}$

(2) 결합법칙 : $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$

◦ 벡터의 뺄셈



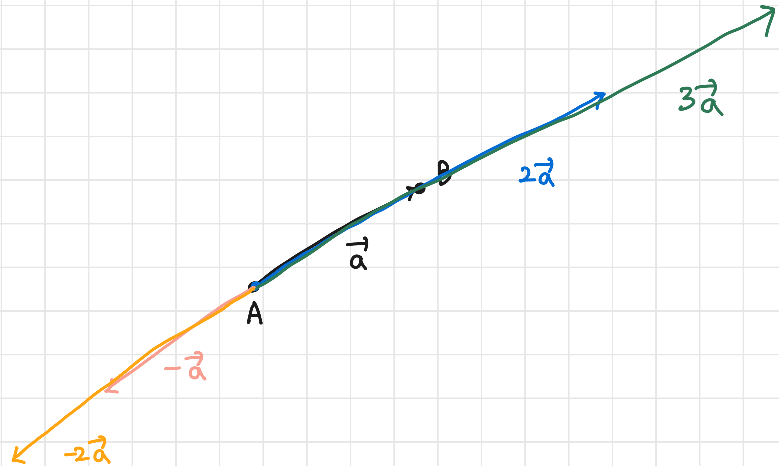
$$\overrightarrow{OB} - \overrightarrow{OC} = (4,3) - (2,3) = (2,0)$$

- 벡터의 스칼라곱 : $k\vec{a}$

(i) $k>0$ 이면, ① 벡터의 크기 $|k\vec{a}|$, ② $\vec{a}$ 같은 방향

(ii) $k<0$ 이면, ② 〃 $|k\vec{a}|$, ② $\vec{a}$ 반대 방향

(iii) $k=0$ 이면, $\vec{0}=(0,0)$



- 평행한 벡터 $\Leftrightarrow$ $\vec{a} \parallel \vec{b}$ $\Leftrightarrow$ $\vec{a} = k\vec{b}$

$\Leftrightarrow$ $d\vec{a} = \vec{b}$

∘ 두 벡터가 평행한 관계라면,

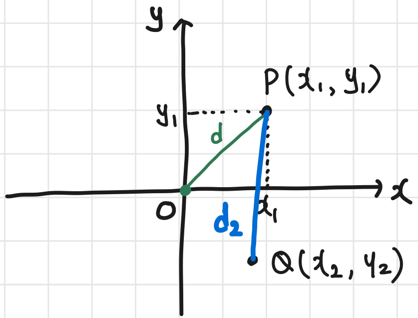
① 평행하다

② 실수배이다

③ 스칼라곱 관계이다

같은 말!

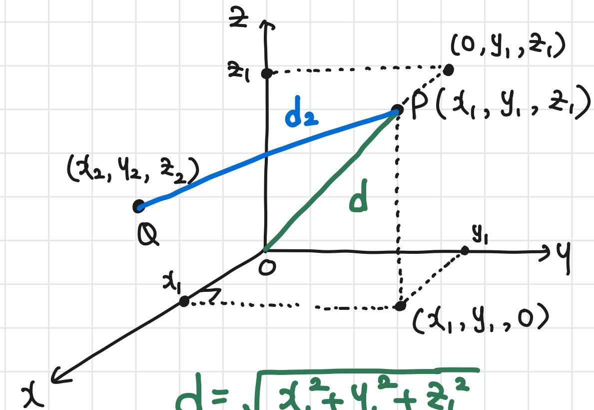
• 평면좌표 (prev) : xy-좌표계



$$d = \sqrt{x_1^2 + y_1^2}$$

$$d_2 = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

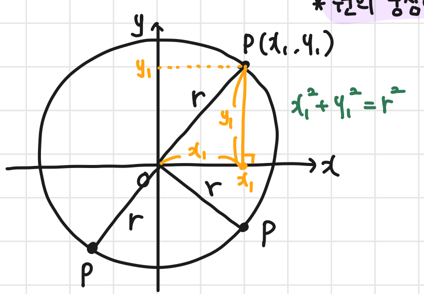
• 공간좌표 : xyz-좌표계



$$d = \sqrt{x_1^2 + y_1^2 + z_1^2}$$

$$d_2 = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

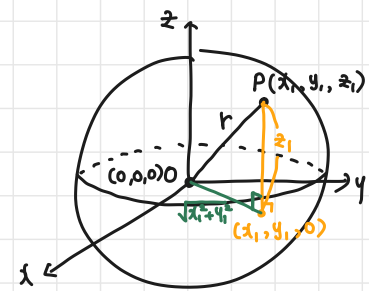
• 원의 방정식



* 원의 중심이 원점 (0,0) 이고, 반지름이 r인 원

$$x^2 + y^2 = r^2$$

• 구의 방정식



* 구의 중심이 (0,0,0) 이고
구의 반지름이 r인 구:

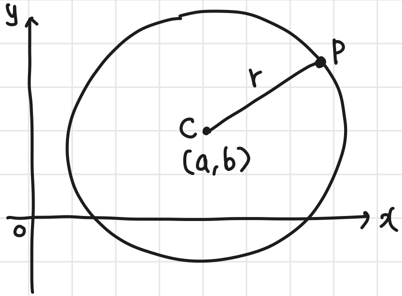
$$z_1^2 + \left(\sqrt{x_1^2 + y_1^2}\right)^2 = r^2$$

$$z_1^2 + x_1^2 + y_1^2 = r^2$$

$$x_1^2 + y_1^2 + z_1^2 = r^2$$

$$x^2 + y^2 + z^2 = r^2$$

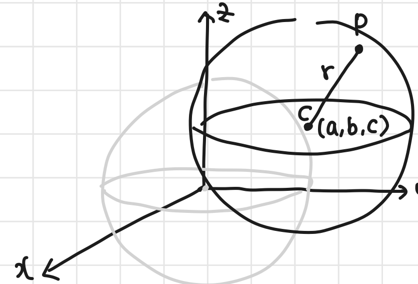
• 원의 방정식



* 중심 (0,0) 반지름 r : $x^2+y^2=r^2$

* 중심 (a,b) 반지름 r : $(x-a)^2+(y-b)^2=r^2$

• 구의 방정식



* 중심 (0,0,0), 반지름 r : $x^2+y^2+z^2=r^2$

* 중심 (a,b,c), 반지름 r : $(x-a)^2+(y-b)^2+(z-c)^2=r^2$

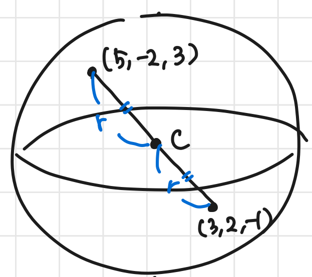
#예 6-10)

(a) 중심 (0,1,2), 반지름 4

$$(x-0)^2+(y-1)^2+(z-2)^2=4^2$$

$$x^2+(y-1)^2+(z-2)^2=16.$$

(b) (5,-2,3), (3,2,-1)을 지름의 양 끝점으로 하는 구.



1. $\left(\frac{5+3}{2}, \frac{-2+2}{2}, \frac{3-1}{2}\right)$

$= (4, 0, 1)$

2. $\sqrt{(5-3)^2+(-2-2)^2+(3+1)^2}$

$= \sqrt{4+16+16} = \sqrt{36}. = 6$

$\therefore r = 3$

$(x-4)^2+(y-0)^2+(z-1)^2 = 3^2$

$\Leftrightarrow (x-4)^2+y^2+(z-1)^2=9$

3. (b) $(3,5,2)$, $(-1,1,-2)$

중심 : $\left(\frac{3-1}{2}, \frac{5+1}{2}, \frac{2-2}{2}\right) = (1, 3, 0)$

반지름 : $\sqrt{(1+1)^2+(3-1)^2+2^2} = \sqrt{4+4+4} = \sqrt{12} = 2\sqrt{3}$

$(x-1)^2+(y-3)^2+z^2 = 12$